

[DC] Supporting Complex Interactions in Mobile Virtual Reality

Kristen Grinyer*

Carleton University, Ottawa, Canada



Figure 1: Left: Low-Fi VR Controller in right hand and cardboard piece with selection markers in left hand depicting the Marker selection method. Right: VR view of the controller with sphere and cube objects indicating tracking statuses of markers.

ABSTRACT

Virtual reality (VR) is poised to become a ubiquitous technology in the coming years. However, its high-cost barrier to adoption prevents user groups of lower socioeconomic statuses from accessing the technology, leaving them out of what may become a predominant computing paradigm in the next decade. Low-cost mobile VR has potential to democratize VR access but offers limited interaction capabilities due to low-fidelity hardware and few input options. Mobile VR cannot currently support most VR applications or offer effective VR experiences. This doctoral consortium proposal outlines my past, current, and future work aiming to investigate and develop low-cost interaction solutions for mobile VR to support complex interactions and outlines topics for future discussion.

Keywords: Virtual reality, pointing devices, mobile VR.

Index Terms: [Human-centered computing]: Human computer interaction—Interaction paradigms—Virtual reality

1 BACKGROUND AND MOTIVATION

In the past decade, advancements in commercial virtual reality (VR) devices and consumer interest have steadily grown¹. Tech giants, such as Meta and Apple are actively developing improved high-fidelity VR head-mounted displays (HMDs) and are pushing for VR and mixed reality (MR) to become an everyday commodity. VR can improve the effectiveness and user experience of applications in work, education, health, and entertainment [3].

The impending ubiquity of VR threatens to widen the digital divide; despite reduced cost, lower socio-economic status (SES) people cannot access the technology and will be left out as VR is established as a new platform for socializing, work, and entertainment. A low-cost version of VR, first introduced in 2014, uses a smartphone encased in a cardboard shell as the hardware. Known as mobile VR (MVR), these low-fidelity HMDs cost significantly less (\$10-20 USD) than modern HMDs (\$300-1400). The mass global adoption of smartphones across income and education levels² is reasonable affirmation users own a smartphone and its cost (\$940 average) is not a factor to MVR access.

MVR allows a larger percentage of the public to access VR technologies [2, 4, 6, 7]. However, a prominent and ongoing issue with MVR is the significantly limited interactions and applications it currently supports [6, 8]. MVR offers few input options; there is

one button on the HMD for input. The smartphone inside the HMD only supports 3DOF head-orientation tracking of the user's head [8]. Translational motion is not supported. Tracking the hands, eyes, or external controllers is not possible without custom software. These tracking limitations prevent MVR from supporting most standard VR interaction techniques and applications. Thus, MVR users (an estimated 135M users worldwide in 2020 [4]) cannot effectively participate in the immersive virtual worlds becoming commonplace nor experience the proven benefits of VR.

Prior MVR research has yielded interaction methods that are notably limited compared to commercial VR [2, 8] or required expensive external hardware [4, 7], contradicting the purpose of MVR by making it less portable and accessible across SES boundaries [8]. The main goal of my doctoral research is to **investigate and develop low-cost MVR interaction methods to support complex interactions** comparable to commercial VR.

2 PRIOR RESEARCH

2.1 Reviewing the Literature

My doctoral research thus far has focused on examining the current state of MVR interaction research; the proposed solutions to improving interaction and their efficacy towards supporting complex interactions in affordable ways. The first milestone of my PhD program I have completed is a literature review and oral exam on the topic of selection in mobile VR. In this review, I extensively researched and wrote on the topics of 1) interaction in 3D environments; 2) selection and manipulation tasks and techniques in VR; 3) evaluating selection performance; 4) known issues of MVR interaction; 5) MVR selection techniques; and 6) a review and analysis of proposed selection techniques to improve interaction in MVR.

My review revealed past research used the inertial sensors of a smartwatch or second smartphone to support 3DOF or 6DOF ray-based selection respectively [4, 7]. The majority of low-cost solutions proposed HMD extensions or gestures on the HMD surface to support menu interaction only [2]. Other solutions used camera-based tracking and hand gestures to support virtual hand and ray-based selection [1, 6]; however, the tracking and gesture recognition software was found unreliable.

My review identified open topics in MVR interaction that past research has not effectively addressed. These include the absence of 1) low-cost and effective 6DOF selection and manipulation techniques; 2) quick and reliable selection indication methods that

* kristengrinyer@email.carleton.ca

¹<https://www.statista.com/statistics/550461/virtual-reality-market-size-worldwide/>

²<https://www.bankmycell.com/blog/how-many-phones-are-in-the-world#part-1>

work universally across selection techniques; and 3) methods to increase camera-based tracking volume to support MVR selection.

2.2 User Study 1

To address these open topics, I designed and evaluated the performance of a low-cost tracked MVR controller and presented this work at SUI'23 [3]. The Low-Fi VR Controller (see Figure 1) uses the smartphone camera to track an external controller with multiple image markers attached to a piece of cardboard. All materials used to make the controller are paper based. Much like commercial VR controllers, the Low-Fi VR Controller provides 6DOF interaction and supports both raycasting and the virtual hand technique. These are the most common selection metaphors used in VR [5, 6], thus it is crucial both are supported in MVR.

The paper [3] details the iterative design process and a pilot study comparing selection confirmation methods. The final design (Figure 1) has three cubic markers with unique images on each side in a triangle formation. The controller pose can be calculated if at least one marker is tracked. The triangle formation effectively increases the tracking boundaries, as the controller can be held farther left, right, or forward compared to having a single marker.

In a formal experiment, I evaluated the efficacy of the controller as a pointing device and compared three selection indication methods including dwell and a novel manual method using a second cardboard marker in the left hand (see Figure 1). This marker-based technique works by tracking the two images (selection markers) on the left-hand cardboard piece; selection occurs when both images are hidden from the camera, i.e., the user flips the marker, akin to pressing a button.

The formal experiment employed the ISO 9241-411 standard selection task to evaluate the controller's selection throughput compared to other pointing techniques. Full details on ISO standard and throughput are seen in the paper [3]. The results indicate the Low-Fi VR Controller offers comparable selection performance to similar ray-based techniques in VR when using dwell to indicate selection. I also recorded qualitative user feedback to identify potential future design improvements.

2.3 Preliminary Work for User Study 2

I am currently updating the controller design to further improve its usability. Study 1 motivates three goals for the next iteration of the controller: 1) reduce tracking jitter, 2) further increase the tracking volume, and 3) create a one-handed solution for manual selection indication. I am currently prototyping one-handed versions of the controller while trying to provide a quick and reliable method for hiding selection markers to indicate selection, without need for a second handheld selection marker.



Figure 2: Early unimanual version of Low-Fi VR Controller

I am currently exploring two concepts (not seen in figures); both involve attaching selection markers to the controller. The first concept employs a slider users can push back and forth to hide selection markers with a piece of cardboard. This is similar to an earlier design (see Figure 2) tested in a pilot study that used a cardboard trigger (with tape to prevent deterioration), but it fell

victim to the Heisenberg effect (unintentional controller movement at the instant of selection). New designs will eliminate this issue.

The second concept uses a UV light source connected to a battery and switch to reveal selection markers printed in UV ink. Proof-of-concept testing revealed that this method can successfully and efficiently show and hide a tracked image. This concept uses a physical button, which is fast and reliable, and low-cost, but requires extra hardware which is not ideal as it makes the design less accessible to MVR users.

3 PLANS FOR FUTURE WORK

I am currently developing two controller prototypes, one for each concept described above, and will evaluate their efficacy as selection indication methods. These results will inform the design of a new controller aimed at addressing the three goals listed in section 2.3. I will then evaluate the performance and user experience of the new controller in the following set of studies.

Future studies will use the same method as the first and will 1) assess the new controller's selection performance with both raycasting and virtual hands; and 2) compare the controller's selection performance to head-gaze selection and a 3DOF Bluetooth controller in MVR. Head-gaze is the most common selection technique for MVR [2, 8] and an important baseline for comparison. Meanwhile, a 3DOF Bluetooth controller offers the closest comparison point to commercial VR controllers. This study will validate improvements of the new controller over the first version and quantify its effectiveness for both ray- and hand-based selection.

After demonstrating the controller's effectiveness as a selection device, I will evaluate the controller's ability to support other fundamental VR interactions: manipulation and travel (viewpoint control) [5], each through its own dedicated user study. All studies mentioned in this position paper will collectively form my PhD thesis. My overarching goal is to design and validate a low-cost input device for MVR to support all fundamental tasks of VR.

At the IEEE VR 2024 doctoral consortium, I would like to discuss the following topics: 1) Quick and reliable methods for changing selection markers between two tracking states to indicate selections; 2) low-cost hardware or software solutions to increase tracking boundaries/camera field of view; and 3) methods to approach object manipulation using the controller.

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